✅ EXPERIMENT 1 – Substitution Ciphers

1. Why is Caesar Cipher insecure?
→ Small key space (only 26 shifts), easy brute force.

2. What is frequency analysis?
→ Technique to break ciphers by analyzing letter frequency.

3. Why is Playfair Cipher stronger than Caesar Cipher?
→ Encrypts pairs (digraphs), increases complexity.

4. What is the difference between monoalphabetic and polyalphabetic ciphers?
→ Mono: one fixed substitution
→ Poly: multiple substitutions (more secure)

5. How can substitution ciphers be broken?
→ Using frequency analysis and pattern recognition.

✅ EXPERIMENT 2 – Transposition Ciphers

a. What is a transposition cipher and how does it differ from a substitution cipher?
→ Rearranges letters without changing them (unlike substitution).

b. Explain the working principle of the Columnar Cipher.
→ Write row-wise, read column-wise using key order.

c. How is the key used in the Columnar Cipher?
→ Decides column reading order.

d. Describe the encryption and decryption process of the Rail Fence Cipher.
→ Zig-zag writing → read row-wise / reverse process.

e. How does the number of rails affect the Rail Fence Cipher?
→ Changes pattern → affects security.

f. Why do transposition ciphers preserve letter frequency?
→ Letters unchanged, only positions change.

g. What are the weaknesses of Columnar and Rail Fence ciphers?
→ Easy to break using pattern + frequency analysis.

h. How can frequency analysis be applied to transposition ciphers?
→ Detect common letters → guess plaintext.

i. Give an example of padding in Columnar Cipher and explain why it is needed.
→ Add 'X' → complete matrix structure.

j. Which cipher is more secure: Columnar or Rail Fence? Why?
→ Columnar (more complex rearrangement).

✅ EXPERIMENT 3 – DES Algorithm

1. What is DES and why is it called a symmetric cipher?
→ Uses same key for encryption and decryption.

2. What is the block size and key size of DES?
→ Block: 64-bit, Key: 56-bit.

3. Explain the difference between ECB and CBC modes.
→ ECB: independent blocks
→ CBC: chained blocks (more secure)

4. Why is ECB mode considered insecure?
→ Same plaintext → same ciphertext.

5. What is the role of Initialization Vector in CBC mode?
→ Adds randomness to first block.

6. Why is padding required in DES?
→ To make data multiple of block size.

7. What is a Feistel structure?
→ Data split → processed in rounds with swapping.

8. How many rounds are used in DES?
→ 16 rounds.

9. Can the same plaintext produce different ciphertext in CBC mode? Why?
→ Yes, due to IV.

10. Why is DES replaced by AES in modern systems?
→ Weak key size, vulnerable to attacks.

✅ EXPERIMENT 4 – RSA Algorithm

1. What is RSA algorithm?
→ Public key cryptography algorithm.

2. What are public and private keys in RSA?
→ Public: (e, n)
→ Private: (d, n)

3. Why are prime numbers used in RSA?
→ Factorization is hard → ensures security.

4. What is the encryption formula in RSA?
→ C = M^e mod n

5. What is the decryption formula in RSA?
→ M = C^d mod n

6. Why is RSA considered secure?
→ Based on difficulty of prime factorization.

✅ EXPERIMENT 5 – Diffie-Hellman

1. What is the main purpose of the Diffie–Hellman key exchange algorithm in secure communication systems?
→ Secure key exchange over insecure channel.

2. Which public parameters must both parties agree upon before starting the Diffie–Hellman key exchange?
→ Prime (p) and generator (g).

3. Why is it computationally difficult for an attacker to derive the shared secret in Diffie–Hellman?
→ Discrete logarithm problem.

4. How is the shared secret generated by both parties without directly transmitting it over the network?
→ Using own private key + other's public key.

✅ EXPERIMENT 6 – Hashing

1. What is a hash function?
→ Converts data → fixed-size hash.

2. Why is MD5 not secure?
→ Collision attacks possible.

3. What is SHA-256?
→ Secure 256-bit hash algorithm.

4. What is collision resistance?
→ Hard to find same hash for different inputs.

5. Why is hashing used?
→ Integrity & authentication.

6. What is the avalanche effect?
→ Small change → big hash change.

7. What module is used in Python for hashing?
→ hashlib

✅ EXPERIMENT 7 – HMAC

1. What is the purpose of HMAC in secure communication?
→ Message authentication + integrity.

2. Why is a secret key required in HMAC but not in a normal hash function like SHA-256?
→ Adds security (not possible in normal hash).

3. What type of attack does HMAC help prevent?
→ Length-extension & forgery attacks.

4. Describe the difference between hashing and message authentication.
→ Hash: integrity
→ HMAC: integrity + authenticity

✅ EXPERIMENT 8 – SSL & Certificates

1. What is a digital certificate?
→ Contains public key + identity.

2. What is SSL/TLS handshake?
→ Process to establish secure connection.

3. Why does a self-signed certificate show a warning?
→ Not issued by trusted CA.

4. What happens during SSL handshake?
→ Key exchange + encryption setup.

✅ EXPERIMENT 9 – Hybrid Cryptography

1. What is hybrid cryptography?
→ Combines symmetric + asymmetric encryption.

2. Why is AES used in hybrid cryptography?
→ Fast for large data.

3. Why is RSA used in hybrid cryptography?
→ Secure key exchange.

4. Explain the working process of hybrid cryptography.
→ Encrypt data with AES → encrypt AES key with RSA.

5. What security features are achieved?
→ Confidentiality, integrity, secure transmission.